HiddenPrompt - Phase 1 Technical Architecture

1. Objective

Define the technical architecture, component responsibilities, game state, communication flow, and project structure before implementation.

2. High-Level Architecture

Client: React + Vite + Tailwind CSS + Socket.IO Client
Server: Node.js + Express + Socket.IO
Core Modules: Game Engine, Room Manager, AI Service
External: AI Image Generation API

3. Why No Database (MVP)

The MVP stores all active game state in server memory (RAM). Since there is no authentication, persistent profiles, or match history, MongoDB is intentionally omitted. A database can be introduced later for leaderboards, statistics, saved matches, and image galleries.

4. Backend Components

Express: Health check and future REST APIs.
Socket Server: Real-time communication.
Game Engine: Round lifecycle, scoring, timers, winner calculation.
Room Manager: Room creation, players, settings, host management.
AI Service: Image generation, regeneration, retries, caching.

5. Frontend Structure

Pages: Landing, Lobby, Game, Results.
Components: PlayerCard, Chat, GuessInput, Timer, Scoreboard, ImageViewer, Settings.
Hooks: useSocket, useGame, useRoom, useTimer.
Context: SocketContext, RoomContext, GameContext.

6. Game State

Global game state includes: roomCode, status, host, players, settings, currentRound, currentPrompter, currentPrompt, currentImage, timer, scores, winner.

7. Data Models

Player: socketId, username, score, isHost, isReady, hasGuessed, connected.
Room: roomCode, hostId, players, settings, game.
Settings: rounds, guessTime, hintInterval, difficulty, maxPlayers.
Round: number, prompter, prompt, image, hintLevel, guessedPlayers, timer.

8. Game State Machine

Waiting → Starting → Prompt Selection → Generating Image → Guessing → Round End → Next Round → Game Over

9. Socket Event Flow

Client emits: create-room, join-room, start-game, submit-guess, ready, next-round.
Server broadcasts: room-created, player-joined, game-started, image-ready, score-update, timer-update, round-ended, game-over.

10. Timer Strategy

The countdown timer is controlled exclusively by the server. Clients only display updates received from the server to ensure synchronization.

11. Image Generation Flow

Prompter selects a prompt → Game Engine requests AI Service → AI API returns image URL → Server broadcasts image to guessing players.

12. Recommended Folder Structure

```
hiddenprompt/
■■■ client/
■  ■■■ pages
■  ■■■ components
■  ■■■ hooks
■  ■■■ context
■  ■■■ services
■  ■■■ types

■■■ server/
■  ■■■ socket
■  ■■■ game
■  ■■■ rooms
■  ■■■ ai
■  ■■■ utils
■  ■■■ types

■■■ shared/
   ■■■ constants
   ■■■ events.ts
   ■■■ types.ts
```

13. Next Steps

Learn Socket.IO fundamentals through small projects (chat, lobby, shared timer), then implement HiddenPrompt using the defined architecture.